JUE Insight: Technological Innovation in Mortgage Underwriting and the Growth in Credit: 1985–2015*

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Abstract

We revisit the evolution of the relationship between mortgage debt and income, adding to the growing literature that extends beyond the 2000s housing boom. Our analysis reveals a permanent decline in the correlation between purchase mortgage amounts and borrower incomes that occurred in the 1990s and was largely complete before the 2000s housing boom. We attribute the decline in the debt-income correlation to a transition in underwriting practices, as lenders moved from human judgment to data-driven statistical models of mortgage default that better identified risky borrowers and enabled broader access to mortgage credit. Using a simple model, we show that patterns observed in the 1990s, including slow price growth and large increases in homeownership, are consistent with a substantial expansion of credit to lower-income households.

Keywords: Mortgage underwriting, housing cycle, technological change, credit boom.

JEL Codes: C55, D53, G21, L85, R21, R31.

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1 Introduction

The mortgage market experienced numerous pivotal changes between 1985 and 2015. The 1990s saw credit scores incorporated into mortgage underwriting, as well as the introduction and widespread adoption of automated underwriting software. The early 2000s saw the expansion of private-label mortgage-backed securities, which allowed for the packaging and sale of subprime and Alt-A loans. Later in that decade came the Great Recession and foreclosure crisis, followed in the 2010s by the growth of fintech. In each of these decades, government policies evolved to facilitate homeownership, and the 30-year mortgage rate experienced a secular decline.

How did these changes affect mortgage underwriting? A common approach for measuring underwriting standards is to examine how mortgage debt rises with income. A steep slope indicates that rigorous lending standards are limiting credit access for low-income borrowers, while a flatter slope suggests greater credit access relative to income. In this paper, we examine the relationship between mortgage debt and income between 1985 and 2015. Our regressions are closely tied to theory and use two sources of borrower-level data: loan information collected pursuant to the Home Mortgage Disclosure Act (HMDA) and data from the American Housing Survey (AHS). Both of these sources include individual levels of debt and income; we complement those sources with tract-level income data compiled by the Census Bureau.

Our regression specifications are motivated by a dynamic equilibrium model of housing and mortgage markets that includes house-spending constraints and endogenous tenure choice. In the model, the debt-income relationship is always positive—richer households tend to have higher debt levels, as they do in the data. However, the model predicts that the debt-income relationship responds differently to different types of changes that may have occurred over the past several decades. Innovations that allow constrained households to increase housing expenditures will flatten the positive debt-income relationship by allowing low-income borrowers to access more credit relative to their incomes. By contrast, changes to user costs for all borrowers—which could arise from changes in interest rates or beliefs about future house prices—lead to parallel shifts in the debt-income relationship, with no change in slope.

Our study makes two contributions. First, we find that the relationship between debt and income did change during our sample period. But this change did not occur during the early 2000s housing boom, as has been widely claimed. Rather, the debt-income relationship flattened in the 1990s; the 2000s boom coincided with a parallel shift up in the debt-income relationship. Through the lens of our model, these patterns suggest that in the 1990s, credit expanded specifically to low-income borrowers. By contrast, the 2000s boom was driven by factors such as lower interest rates or widely held optimistic beliefs about the housing

market. These factors increased mortgage debt across all income levels, shifting the debt-income relationship up without changing its slope.

Mian and Sufi (2009) also studied the debt-income relationship during the housing boom, but our findings contradict theirs because of some key methodological differences. First, as pointed out by Adelino et al. (2016), Mian and Sufi (2009) focused on the total amount of mortgage debt originated in a zip code, rather than the amount of debt that a borrower with a given income could expect to receive. This choice conflates the extensive and intensive margins of a potential credit expansion, which are best studied independently with individual-level data, as we do below. Second, Mian and Sufi (2009) looked at *first differences* of debt and income data calculated at the zip-code level, rather than at *levels* of income and debt among individual borrowers. Because aggregating to the zip-code level and taking first differences throws away a great deal of identifying variation, the approach generates unstable estimates of the debt-income slope that are attenuated towards zero and may even be negative.[1]

Our finding of stability in the debt-income slope during the 2000s is consistent with the results of Adelino et al. (2016) and is robust to two data concerns raised later by Mian and Sufi (2017): alternative definitions of income and proper treatment of second liens. Incomes on loan applications in HMDA data can be overstated if borrowers are stretching to qualify for mortgages. But income data collected by the Census and the AHS, which are used in our baseline results, do not suffer from this problem. Additionally, researchers using public versions of the HMDA data cannot match first liens to second liens within the same households before 2004. We use a confidential version of the HMDA data that allows such matching in all years.

The finding of a uniform increase in debt during the 2000s supports the "new narrative" of the boom that was sparked by Adelino et al. (2016) and also includes Albanesi et al. (2022), Kaplan et al. (2020), Conklin et al. (2022), Ferreira and Gyourko (2015), Foote et al. (2012, 2021), Gennaioli and Shleifer (2018), and others. This narrative claims that the 2000s boom was driven by excessive optimism about future house price appreciation—which lowers user costs for all households—not by an exogenous relaxation of underwriting standards that favored low-income borrowers.[2]

Our second contribution is to explain why the debt-income relationship changed in the 1990s. We argue that this period marked a transformation in underwriting practices. Lenders shifted from a traditional judgmental approach that relied on the intuition and experience of underwriters to a modern, data-driven statistical approach. Similar transformations had

[1] In the appendix, we display this instability for the years that Mian and Sufi (2009) studied.

[2] Our earlier paper, Foote et al. (2021), investigates credit-bureau data on the *stock* of mortgage debt across households during the housing boom. This paper measures the *flow* of new mortgage debt—which is directly related to underwriting standards—over a much longer period.

already taken place in other areas of consumer lending, including credit cards. Mortgage lenders initially resisted this change, arguing that human intuition remained essential for predicting mortgage default. But once lenders started using the statistical approach, it proved far superior to human judgment, just as it had for other types of consumer debt.

The adoption of statistical techniques for mortgage underwriting was part of a transformation of the analysis of human behavior that stretched over much of the 20th century (Meehl 1954). Research has shown that simple statistical models typically outperform human judgment for a wide range of predictions, such as which students will succeed in graduate school, which persons will suffer heart attacks, and which former inmates will violate their paroles (Dawes et al. 1993). Advanced forms of statistical analysis, including machine learning, continue to expand the set of questions for which machines outperform humans.

The link between our empirical findings and the adoption of statistical techniques arises because one of the key findings from statistical analysis of mortgage data is that human underwriters placed excessive weight on labor income when deciding whether to approve a loan. Early empirical models showed that once credit history is accounted for, the payment-to-income ratios favored by underwriters had limited predictive power.[3] This fact may not surprise labor economists. As early as MaCurdy (1982), researchers demonstrated that labor income shocks were larger and more persistent than previously understood. Consequently, current income provides limited insight into a borrower's ability to meet financial obligations many years in the future, the relevant horizon for forecasting mortgage default.

After establishing that underwriting practices changed in the 1990s, we use our theoretical model to study the effects of this change on the homeownership rate and house prices. For homeownership, the model makes an unambiguous prediction about the effect of financial innovation: homeownership should increase. This prediction is confirmed in the data, as there was a sharp increase in the homeownership rate from 1995 to 2001, concentrated in low-income households, that was concurrent with the change in the relationship between mortgage debt and income.

The model's implication for house prices is less clear-cut. One might expect innovation to raise prices by increasing demand both intensively (greater spending among current owners) and extensively (more households buying homes). However, our model implies that the extensive margin can actually *reduce* demand. This finding depends on two assumptions: first, that households prefer to own rather than rent, and second, that rental housing and owner-occupied housing are substitutes. The former assumption implies that marginal

[3] An early Federal Reserve study (Avery et al. 1996) summarized the existing consensus in both industry and academia by noting that "[p]erhaps surprisingly, after controlling for other factors, the initial ratio of debt payment to income has been found to be, at best, only weakly related to the likelihood of default" (p. 624). Recent research has shown that payment-to-income ratios at origination are more helpful in predicting default than initial research had suggested. But modern models of mortgage default still find more powerful roles for loan-to-value ratios and credit scores. See Davis et al. (2022) for a discussion.

households will accept smaller houses when moving from renting to owning, reducing the aggregate demand for housing. The substitutability assumption implies that this reduced demand puts downward pressure on both prices and rents.

Are these assumptions reasonable? Survey data overwhelmingly support the idea that most households prefer to own rather than rent.[4] Regarding substitutability, our analysis of AHS data shows that the increase in homeownership during the 1990s was facilitated by the conversion of rental properties into owner-occupied homes. This pattern stands in contrast to the usual pattern of owned properties becoming rentals as they age (Rosenthal 2014). But a reversal of the usual pattern is consistent with our model if rented and owned homes are substitutes and if an innovation has relaxed debt-income constraints.

Our paper contributes to the growing body of work on how financial innovation affects mortgage markets. A closely related paper is by Johnson and Tzur-Ilan (2024), who study the effect of automated-underwriting adoption. They use a difference-in-differences approach that takes advantage of the relative sophistication of the automated underwriting software created by Freddie Mac versus that of Fannie Mae. They find that areas where lenders had a stronger pre-existing relationship with Freddie Mac saw greater house price growth, indicating that an expansion of credit due to the adoption of automated underwriting increased house prices. By contrast, we show that real house prices did not start to rise rapidly until much later. Indeed, real house prices increased only minimally in the 1990s.

Another related paper is by Davis et al. (2022), who assemble data from Fannie Mae, Freddie Mac, private-label securitizers and other sources to study the evolution of several mortgage-risk characteristics from 1990 to 2019. Data limitations prevent them from analyzing payment-to-income ratios before 1993 and credit scores before 1994. But the authors find a general relaxation of credit in the 1990s, before the housing boom began. Developments in the 1980s also expanded credit. Gerardi et al. (2010) find that the increased use of securitization and alternative mortgage products in that decade strengthened the relationship between home purchase value and *future* household income. The implication is that relaxed credit markets improved households' ability to smooth consumption by allowing them to borrow more against future income.[5]

Section 2 uses our model to illustrate the implications of financial innovation on the housing market. We present our empirical findings in section 3, and section 4 relates these findings to historical changes in mortgage underwriting. Section 5 concludes.

[4]The New York Fed Survey of Consumer Expectations shows that, all else equal, less than 20 percent of households prefer to rent rather than own. See https://www.newyorkfed.org/microeconomics/sce/housing.

[5]Other papers in this literature include Fuster et al. (2022, 2019), Philippon (2019) and Rudolph and Griffith (1997).

2 Three Observations from a User Cost Model with Constraints

This section presents a simple model to show how underwriting innovations that expand credit can affect the housing market. Our model is a standard infinite-horizon continuous-time framework in the spirit of Poterba (1984) and Hornstein (2009).

2.1 The Household Problem

Consider an infinitely lived representative household with $N_t = N_0 e^{nt}$ members, where n is the population growth rate. The household derives utility from per capita consumption c_t and housing h_t, with preferences that depend on tenure status. The household maximizes lifetime utility:

$$\max \int_0^\infty e^{-\rho t} N_t \log\left(c_t^{1-\theta}(h_t/N_t)^\theta \omega^{I_t}\right) dt.$$

Here, $\rho > 0$ is the subjective discount rate, h_t is the household's housing stock, $\theta \in (0,1)$ represents the expenditure share on housing services, and I_t is an indicator variable equal to 1 if the household owns and 0 if it rents. The parameter ω measures individual household preferences for owning versus renting and takes any positive value ranging from zero to infinity from a smooth distribution F, where $F(x) = P(\omega < x)$. All else equal, if $\omega = 1$, the household is indifferent between owning and renting. If $\omega > 1$ ($\omega < 1$), it prefers to own (rent).

A dynamic asset accumulation equation and a no-Ponzi-game condition imply a lifetime budget constraint for owners:

$$\int_{s=t}^\infty e^{-R(t,s)} \left[N_s c_s + \underbrace{P_s^h \left(r_s + \tau + \delta - \dot{P}_s^h / P_s^h\right)}_{=\mathrm{uc}_s} h_s \right] ds = a_t + P_t^h h_t + \int_{s=t}^\infty e^{-R(t,s)} N_s y_s ds,$$

where P_s^h is the price of a unit of housing at time s, uc_s denotes the user cost of housing at time s, $R(t,s) = \int_t^s r_u du$ is the cumulative discount factor, r_s is the risk-free real interest rate, a_t is financial wealth, and $y_t = y_0 e^{gt}$ is per capita labor income. Two constant parameters, τ and δ, denote a proportional real estate tax and the depreciation rate of housing, respectively. Replacing the user cost uc_s with rent_s yields a similar budget constraint for renters.

We model credit constraints by imposing a flow limit on how much a household can spend on housing relative to income:

$$\text{(a) } \mathrm{rent}_t \cdot h_t \leq \overline{\theta}_r N_t y_t \qquad \text{(b) } \mathrm{uc}_t \cdot h_t \leq \overline{\theta}_o N_t y_t, \tag{1}$$

where $\overline{\theta}_r$ and $\overline{\theta}_o$ are institutional parameters governing the maximum housing expenditure shares for renters and owners, respectively.

The household's optimal choices satisfy the following first-order conditions. For consumption, the Euler equation is:

$$\frac{\dot{c}_t}{c_t} = r_t - \rho.$$

For unconstrained households, the optimal allocation between housing and consumption for owners satisfies:

$$\frac{\mathrm{uc}_t \cdot h_t}{N_t c_t} = \frac{\theta}{1-\theta},$$

with an equivalent condition for renters that replaces uc_t with rent_t.

The tenure choice depends on a comparison of utility under owning and renting. A household with preference parameter ω chooses to own if the utility from owning exceeds that from renting, given optimal consumption and housing choices under each tenure mode.

Finally, investors initially hold the housing stock and derive utility from consumption. They can either (1) own housing and lease it, or (2) sell housing to households and invest the proceeds in the riskless asset.

2.2 Balanced Growth Path

In equilibrium, there is a balanced growth path with:

$$\text{(a) } r_t = r = \rho + g \qquad \text{(b) } \frac{\dot{P}_t^h}{P_t^h} = g + n,$$

which implies that:

$$\mathrm{uc}_t = P_t^h \left(r_t + \tau + \delta - \frac{\dot{P}_t^h}{P_t^h} \right) = P_t^h (\rho + \tau + \delta - n). \tag{2}$$

Along the balanced growth path, we can solve for housing demand for unconstrained households:

$$\text{(a) } \mathrm{uc}_t h_t = \theta N_t y_t \quad \text{(owners)} \qquad \text{(b) } \mathrm{rent}_t h_t = \theta N_t y_t \quad \text{(renters)}. \tag{3}$$

The condition that $\mathrm{uc}_t = \mathrm{rent}_t$ is ensured by (1) the substitutability of renter and owner-occupied housing and (2) the ability of investors to either buy properties and rent them out or sell properties to households and invest the proceeds in bonds earning the risk-free rate.

We partition the population by financial constraints: a fraction $1-\alpha$ is unconstrained (with $\theta \leq \min\{\overline{\theta}_o, \overline{\theta}_r\}$) and a fraction α is constrained (with $\theta > \max\{\overline{\theta}_o, \overline{\theta}_r\}$). Unconstrained households own if and only if $\omega > (\mathrm{uc}/\mathrm{rent})^\theta$. For constrained households, equations (1) (a)

and (b) hold with equality, implying that the threshold $\overline{\omega}$ for constrained households is:

$$\overline{\omega} = \left(\frac{1-\overline{\theta}_r}{1-\overline{\theta}_o}\right)^{1-\theta} \left(\frac{\overline{\theta}_r}{\overline{\theta}_o}\right)^{\theta} \left(\frac{\text{uc}}{\text{rent}}\right)^{\theta} = \left(\frac{1-\overline{\theta}_r}{1-\overline{\theta}_o}\right)^{1-\theta} \left(\frac{\overline{\theta}_r}{\overline{\theta}_o}\right)^{\theta}. \tag{4}$$

To link constraints on housing spending to income, we further assume that low-income households are more likely to have high values of θ, so they are more likely to be constrained.[6]

We can now study the effect on market equilibrium:

$$\underbrace{\left[\alpha\left((1-F(\overline{\omega}))\overline{\theta}_o + F(\overline{\omega})\overline{\theta}_r\right) + (1-\alpha)\theta\right]\frac{N\cdot y}{uc}}_{\text{Demand}} = \underbrace{\overline{h}}_{\text{Supply}}. \tag{5}$$

In this expression, the term multiplying α reflects the demand contribution of constrained households. The θ that multiplies $(1-\alpha)$ reflects the spending preferences of unconstrained households, who are assumed to share the same value of θ.

2.3 Effect of a Financial Innovation

We next consider a financial innovation that relaxes credit constraints and allows borrowers to allocate a bigger fraction of their income to housing. Formally, we model this as an increase $\overline{\theta}_o$. The model yields three main empirical predictions:

(1) Financial innovation affects the slope of the relationship between income and debt, while changes in user cost shift this relationship without affecting its slope. This distinction arises because credit restrictions bind only for low-income households, whereas user cost changes affect all households uniformly. To illustrate this mechanism, consider two households with incomes $Y_1 > Y_2$. Household 1 is unconstrained, while Household 2 faces a binding credit constraint on housing. Combining equations (2), (1) and (3) yields their respective housing demands:

$$\text{(a) } P^h h_1 = \frac{\theta N Y_1}{\rho+\tau+\delta-n} \qquad \text{(b) } P^h h_2 = \frac{\overline{\theta}_o N Y_2}{\rho+\tau+\delta-n}. \tag{6}$$

Figure 1 depicts the respective household decisions graphically, with the baseline scenario (black line) showing that Household 1 spends more on housing than Household 2, reflecting both its higher income ($Y_1 > Y_2$) and its freedom from credit constraints ($\theta > \overline{\theta}_o$). Now imagine one of two shocks hit the economy. First, a credit expansion (increasing $\overline{\theta}_o$) allows Household 2 to increase housing spending from h_2 to h_2', while Household 1's spending remains unchanged since it was never constrained. This selective impact on constrained households flattens the income-housing relationship (red line in Figure 1). Second, a user cost

[6] As we will show in our empirical work, the slope of the empirical debt-income gradient is positive but less than 1. This fact implies that as household income rises, the desired spending share on housing falls.

reduction shifts the entire relationship upward (blue line in Figure 1) because both households face the same denominator in their demand equations. Importantly, the mechanisms differ: unconstrained Household 1 directly responds to cheaper housing services, while constrained Household 2 benefits indirectly through lower mortgage payments that create room within its existing credit limit.

(2) Financial innovation increases homeownership. To illustrate this prediction, consider the marginal renter who is indifferent between owning a house of size h_o and renting a house of size h_r. When we relax the ownership constraint $\overline{\theta}$, the marginal household can now afford to own a house of size $h_o^{'} > h_o$ and now strictly prefers to own, increasing the rate of homeownership. Formally, differentiating equation (4) with respect to $\overline{\theta}_o$ yields

$$\frac{\partial \overline{\omega}}{\partial \overline{\theta}_o} = \left(\frac{1-\overline{\theta}_r}{1-\overline{\theta}_o}\right)^{1-\theta} \left(\frac{\overline{\theta}_r}{\overline{\theta}_o}\right)^{\theta} \left(\frac{1-\theta}{1-\overline{\theta}_o} - \frac{\theta}{\overline{\theta}_o}\right) < 0 \Leftrightarrow \overline{\theta}_o < \theta. \tag{7}$$

(3) Financial innovation has ambiguous effects on prices and rents. One might expect credit expansion to raise prices by increasing demand both intensively (higher spending among current owners) and extensively (more households buying homes). However, the extensive margin can *reduce* demand. To see this, differentiate the left-hand-side of equation (5) with respect to $\overline{\theta}_o$ to get:

$$\tfrac{\partial \text{Demand}}{\partial \overline{\theta}_o} \quad = \quad -\alpha F'(\overline{\omega}) \tfrac{\partial \overline{\omega}}{\partial \overline{\theta}_o} \left(\overline{\theta}_o - \overline{\theta}_r\right) \cdot \tfrac{Ny}{\text{uc}} + \alpha\big(1 - F(\overline{\omega})\big) \cdot \tfrac{Ny}{\text{uc}}. \tag{8}$$

The first term on the right-hand side measures the net effect of tenure transitions. By equation (7), $-F'(\overline{\omega})\frac{\partial \overline{\omega}}{\partial \overline{\theta}_o} > 0$, so the sign of the first term depends on the relative sizes of the ownership and renting constraints. If the ownership constraint is tighter than the renting constraint ($\overline{\theta_o} - \overline{\theta_r} < 0$), then transitions to ownership reduce demand. To understand why, note that for the marginal renter, the benefits of a bigger house exactly offset the warm glow of homeownership. Relaxing the ownership constraint tilts the balance and the marginal renter transitions to a smaller house. This transition to homeownership raises the former renter's utility but reduces the aggregate demand for housing. The second term represents the existing constrained homeowners who now can spend more.

Two points about the model are worth noting. First, equation (6) shows that house prices increase when interest rates fall (lower ρ) or when expected price growth rises (higher n). Because these movements affect user costs, they affect both constrained and unconstrained households equally. Second, if we relax the assumption that owner-occupied and rental housing are substitutable, then the extensive margin effect can increase house prices as more households compete for a fixed supply of owner-occupied homes. However, this effect

is attenuated by unconstrained owners who transition to renting as the relative cost of homeownership increases.

3 Empirical Evidence of a Credit Expansion

3.1 Mortgage Debt and Income in the HMDA Data and the AHS

We first study the relationship between income and mortgage debt using data from HMDA from 1990 through 2015. With some caveats discussed below, these data include individual mortgage amounts, each mortgage's purpose (purchase or refinance), whether the purchased property will be owner-occupied, and household income. We limit our attention to owner-occupied purchase mortgages. Our use of confidential HMDA data allows us to combine amounts from both first and second liens originated at purchase, which is not possible with public-use HMDA data before 2004. The online appendix provides the details of our accounting for second liens along with additional information about how we clean the data.

It is well known that incomes in HMDA data are often misreported (Avery et al. 2012; Blackburn and Vermilyea 2012; Mian and Sufi 2017). Underreporting occurs when a borrower reports only the income needed to qualify for a loan; overreporting occurs when borrowers inflate incomes to meet qualification requirements. We address this problem by instrumenting for borrower income using median household income by tract from decennial censuses and the American Community Survey (ACS).[7]

Using the HMDA data we run the following regression:

$$D_{it} = \alpha_{ct} + \beta_t I_{it} + \gamma X_{it} + \epsilon_{it}, \tag{9}$$

where D_{it} is the combined value of purchase mortgages for borrower i in year t, I_{it} is household income, X_{it} includes dummy variables denoting the borrower's race and gender, and c indexes the core-based statistical area (CBSA) where the property is located. Note that α_{ct} is indexed by CBSA as well as by time, so this term represents CBSA-year fixed effects. Standard errors are clustered by CBSA.

The estimated β_ts from this regression illustrate how the relationship between debt and income changed over time. In Figure 2a, these coefficients trend downward from 1990 to the start of the housing boom in the early 2000s. The coefficients then remain relatively flat through the end of our sample in 2015.

Figure 2b presents estimates from quantile regressions at the 25th, 50th, and 75th per-

[7]Results obtained without instrumenting appear in Figure A.2 in the appendix. Figure A.3 in the appendix includes a comparison of our baseline results to results when second liens are not identified and matched to first liens.

centiles of the income distribution. These regressions are similar to Equation (9), except that the quantile models omit controls for race and gender. Lower-income individuals generally have higher slope coefficients, indicating that lenders tie income more closely to debt for these borrowers.[8] During the 1990s, the 25th-percentile coefficients decline by more than coefficients at higher percentiles, indicating a greater relaxation of credit for lower-income households. During the Great Recession, the 25th-percentile coefficient rises slightly, consistent with a tightening of mortgage credit for low-income borrowers as the housing market declined (Goodman 2017). Yet all coefficients remain substantially below their early-1990s values.

Figure 2c depicts the predicted values of the OLS regression after omitting CBSA-fixed effects α_{ct} and assuming the mean income of the entire sample. Expected mortgage amounts increase sharply after 2000, when the well-documented increase in the ratio of aggregate mortgage debt to aggregate income takes place. Combined with the stable income coefficients over the same period, this pattern indicates that mortgage originations rose proportionately across the income distribution during the housing boom.

We also use the AHS to test the debt-income relationship over time. Household income in the AHS is not reported to qualify for loans, so we do not need to instrument for income, and first and second liens are much easier to combine. Disadvantages of the AHS include its smaller size and the fact that mortgage amounts come from survey responses, not administrative data.[9] We use the AHS to run a version of Equation (9) that excludes CBSA fixed effects due to the small sample size and present the results in Figures 2d through 2f. A comparison of each AHS figure with the HMDA figure immediately above it displays strong agreement between the two data sources.

3.2 Predictions for Homeownership Rates and House Prices

Our model predicts that a credit expansion raises the homeownership rate among previously constrained households. To test this hypothesis, we use the AHS to calculate homeownership rates for all households with heads aged 25–64, as well as for five income-based quintiles from this group.[10] The thick black line in Figure 3a shows between 1989 and 2001, the overall homeownership rate rose 5.4 percentage points from the 1989 rate of 63.5%. Gains were particularly large among the lowest three income quintiles, with increases of 6.5 points, 5.8

[8] Figure A.2 in the appendix displays quantile regressions that do not instrument for borrower income with census-tract income. In these regressions, income coefficients for the highest-income borrowers are largest, possibly indicating that high-income borrowers have more flexibility to adjust their reported incomes on mortgage applications.

[9] See the online appendix for a fuller discussion of AHS data.

[10] The aggregate homeownership rate generated by the AHS coincides closely with the official rate generated by the Current Population Survey–Housing Vacancy Survey (CPS-HVS). See the appendix for details.

points, and 6.8 points for the bottom, second, and third quintiles, respectively. During the early-2000s boom, the overall rate flattens out, as modest increases in higher-income groups are offset by a decline in the lowest-income group. Specifically, the rate for the lowest quintile declines from 41.1% in 2001 to 37.1% in 2007, and rates for the second and third quintiles rise by just over 1 percentage point each over this period. These patterns are consistent with our model, which predicts increased homeownership for lower-income groups when the debt-income gradient flattens. There is no significant increase (and a decline among the lowest group) during the boom, when gradient undergoes a parallel shift up.

The model predicts that a credit expansion has an ambiguous effect on house prices, due to potentially offsetting extensive- and intensive-margin effects. Figure 3b plots annualized real house price growth by census region from 1990 to 2000 and from 2000 to 2007. In the 1990s, aggregate real house prices were flat, with some variation across census regions. In the 2000s, however, prices grew across all regions.

The key assumption that allows the model to be consistent with slow price growth during a credit expansion is that the housing stock is substitutable between renters and owner-occupiers (Rosenthal 2020). In the model, when a household transitions from renter to owner, it may, at least theoretically, continue to occupy the same house, resulting in little price response as supply continues to meet demand.

We can see this substitution occurring in the AHS, which follows individual housing units over time. Units often switch from being rented in one wave of the survey to being owner-occupied in the next (and vice versa). With appropriate sample weights, the AHS can be used to calculate the number of units that transition from one form of tenure to another.[11] Results of this exercise for owner-occupied homes appear in Figure 3c. The dark blue bars depict inflows to the owner-occupied housing stock from new construction. The red bars are net additions from the rental housing stock, the gray bars are net additions from vacant properties, and the pink bars are net additions from seasonal properties. Most of the time, the red bars lie below the horizontal axis, indicating that it is more common for an owner-occupied property to become rented than vice versa. Indeed, research on the filtering of aging owner-occupied properties to rentals describes precisely this pattern (Rosenthal 2014). Yet from 1997 through 2001, net additions from the renter-occupied housing stock became positive.

The late-1990s switch from traditional filtering to the net conversion of rental properties to owner-occupied ones illustrates two points. First, our modeling assumption that the housing stock is substitutable is valid, and second, the late 1990s saw an increase in demand for owner occupancy relative to previous periods.

[11] We based our weighting procedure on the method described in Eggers and Moumen (2005). The appendix explains the relationship of our analysis to the Department of Housing and Urban Development's Components of Inventory Change (CINCH) report, which is also generated with AHS microdata.

3.3 Relation to Previous Research

In a widely cited paper, Mian and Sufi (2009) found a negative correlation between mortgage credit growth and income growth at the zip-code level from 2002 to 2005, suggesting that credit flowed disproportionately to lower-income areas. This finding generated considerable debate and appears to contradict our results showing a positive relationship between debt and income in each year of our sample period.

The key to resolving this contradiction lies in recognizing what Mian and Sufi actually measured. As Adelino et al. (2016) demonstrate, Mian and Sufi's dependent variable—total mortgage origination volume—combines the number of loans originated (extensive margin) and average loan size (intensive margin). When decomposed, their negative correlation was driven almost entirely by the extensive margin: more mortgage transactions occurred in lower-income zip codes, while the relationship between average loan sizes and borrower income remained positive—consistent with our individual-level findings.

These differences raise a question: did the increased transaction volume in lower-income areas that Mian and Sufi (2009) identified represent a credit expansion along the extensive margin, which our focus on the debt-income relationship has missed? Or did the higher volume instead reflect higher turnover among existing homeowners in lower-income communities? Figure 3a provides evidence against an extensive-margin expansion. Significant homeownership gains among low- and middle-income groups ended in 2001 and, in the case of the lowest income group, homeownership actually fell during the boom years.[12]

The stability of homeownership rates in the 2000s suggests that the increased transaction volume in lower-income zip codes reflected increased churn—more frequent buying and selling among existing homeowners—rather than expanded homeownership to previously excluded borrowers. This interpretation aligns with documented increases in speculative buying and flipping during the boom. For example, Bayer et al. (2020) show that more than 15% of all Los Angeles homes purchased near the peak of the housing market boom (2003–2005) were resold within two years (see also Haughwout et al. (2011)).

A second methodological difference is that Mian and Sufi (2009) used first-differenced specifications, which create several econometric problems: (1) first-difference specifications suffer from omitted variable bias when coefficients are time-varying (2) they can lead to severe attenuation bias in the presence of transitory income variability and measurement error, and (3) they require the use of aggregate data, which leads to composition issues. We explore these issues in more detail in the appendix, where Figure B.7 shows that differenced specifications do in fact lead to implausibly low and unstable estimates of the debt-income relationship.

[12] In Foote et al. (2021), we show that there were higher rates of mortgage churn (higher originations and higher terminations) in low-income areas during the boom.

4 Developments in Underwriting Technology in the 1990s

In this section, we examine the introduction of automated underwriting, which we argue is the most likely explanation for the decline in the slope of the debt-income relationship in the 1990s. One reason for this view is that adoption of automated underwriting between 1995 and 2000 aligns closely with the period that the debt-income relationship changed. Second, a review of the historical record indicates that automated underwriting significantly expanded credit access.

The shift from manual to automated underwriting marked a fundamental transition from judgmental to statistical credit evaluation (Board of Governors of the Federal Reserve System 2007). Historically, mortgage credit decisions were largely based on the subjective assessments of loan officers, who relied on personal judgment, experience, and sometimes intuition to assess the creditworthiness of applicants (Pease and Cherrington 1953). This method remained the dominant form of mortgage underwriting until the early 1990s (Straka 2000). The introduction of automated underwriting systems in the mid- to late 1990s replaced human discretion with statistical models that analyzed large volumes of historical data to assess risk more objectively and consistently. To calculate the likelihood of repayment, the models incorporated a range of variables, including credit history, income, debt levels, and employment history.

An early example of automated risk assessment illustrates how automated underwriting expanded credit. In September 1994, Freddie Mac developed the Gold Measure Worksheet to evaluate loans for its Affordable Gold program.[13] The worksheet assigned "risk units" to loan applications based on factors such as the borrower's loan-to-value (LTV) ratio, payment-to-income ratio (commonly called a debt-to-income ratio, or DTI), credit score, and other information. Statistical default modeling informed how many risk units each characteristic represented. Accordingly, the worksheet prioritized LTVs and credit scores, which strongly predicted default, and placed less importance on DTIs, which were weaker predictors. Loans with a total risk-unit score of 15 or lower met Freddie Mac's underwriting standards. We reprint this worksheet as Figure 4.

Table 1a illustrates the prioritization of LTVs and credit scores by using the worksheet to evaluate three hypothetical loans. Loan A has an LTV of 90 percent, a credit (FICO) score exceeding 790, and a DTI ratio exceeding 50.5 percent. The high FICO score offsets the high DTI ratio by enough to reduce the total risk-unit score to 14. Loan B has a low

[13] The Affordable Gold program was the company's response to the Federal Housing Enterprises Financial Safety and Soundness Act of 1992, which encouraged the government-sponsored enterprises to increase mortgage credit availability to underserved areas. See Avery et al. (1996) and Straka (2000) for additional discussion of the Affordable Gold program and the Gold Measure Worksheet.

credit score and carries a much lower DTI ratio, but with a risk-unit score of 17, it turns out to be too risky. Finally, Loan C has very high LTV and DTI ratios (99.5 percent and 50.5 percent, respectively), as well as an adjustable interest rate. But these are offset by a credit score of 790 and five months of liquid financial reserves, which give it a risk score of 15.

The Gold Measure Worksheet represented a shift from the previous judgmental approach to underwriting because it quantified and standardized the tradeoffs between different risk factors based on the historical correlations between these factors and actual default rates. Without the worksheet, one underwriter might categorize Loan A as too risky due to its high DTI ratio, while another underwriter might assert that it was eligible for purchase by Freddie Mac. But these assessments would be based on each underwriter's intuition and often limited experience with loans that ended in foreclosure, not on large-scale data analysis.

By 1995, both of the large government-sponsored enterprises (GSEs) had extended the approach illustrated by the Gold Measure Worksheet by developing their own automated underwriting systems: Freddie Mac's Loan Prospector and Fannie Mae's Desktop Underwriter. These systems enabled loan originators to input borrower and loan data and obtain an automated decision on whether the loan met GSE standards. For example, if Loan Prospector gave a loan an "accept" rating, Freddie Mac would purchase the loan without further analysis, while "caution" or "refer" ratings indicated the need for more scrutiny.

Automated underwriting proved to be far more accurate than human underwriters, so it allowed the GSEs to increase their acceptance rates without increasing their risk exposure. This is most directly illustrated by a series of horse races using a sample of loans from the Affordable Gold Program that Freddie Mac purchased in 1994 and reevaluated using both human underwriters and the Gold Measure Worksheet. Because Freddie Mac owned these loans, it could follow them over time to ascertain their true default rates. Later, it would evaluate the loans again using two versions of its automated underwriting software.

Table 1b presents statistics from these evaluations obtained from articles written by Freddie Mac researchers. There are two main takeaways. The first is that statistical evaluation of loans was far more effective than human judgment. The first row of the table shows that manual underwriters accepted 51.6% of the loans for Freddie Mac's backing. But the eventual default rates for their accepted and rejected loans were effectively identical; in fact, defaults were slightly higher for the loans that the human underwriters accepted.[14] In other words, human underwriters' assessments were no better than coin flips at selecting loans that would not default (Straka 2000).

The second row shows that the Gold Measure Worksheet did much better. It accepted about half the loans (like the underwriters) but the loans selected by the worksheet defaulted

[14]Default rates are relative to the average for the entire sample, so they should be interpreted relative to each other, not as raw delinquency rates.

at only 18 percent of the overall sample rate. The delinquency rate for its rejected loans was nearly two times greater than the sample average. The third row shows that the 1995 version of Freddie Mac's Loan Prospector performed effectively the same as, if not slightly worse than, the Gold Measure Worksheet. But these results are still striking. Although both human underwriters and the data-based approaches accepted roughly the same proportion of loans, using data proved far superior to using human judgment at identifying loans with high default probabilities.

The second takeaway from Table 1b concerns acceptance rates. Manual underwriters determined that slightly more than half the loans were acceptable for purchase by Freddie Mac, and the Gold Measure Worksheet had a similar acceptance rate. The 1995 version of Loan Prospector, in addition to performing slightly worse than the worksheet in terms of default rates, also had a lower acceptance rate, about 45 percent. But the acceptance rate of the 2000 version of Loan Prospector exceeded 87 percent. As the GSEs gained confidence in their underwriting software, they realized that the improved precision of the underwriting decisions meant that they could increase their acceptance rates without increasing their aggregate exposure to credit risk (Gates et al. 2002). The default rate on loans accepted using the 1995 version of Loan Prospector was, in a sense, too low. Freddie Mac could accept more marginal loans, which led to default rates that were higher but still within acceptable limits.

5 Conclusion

In this paper, we develop a theoretical model that generates three empirical implications of an exogenous mortgage credit expansion: a decline in the partial correlation between income and mortgage debt, an increase in the homeownership rate, and an ambiguous effect on real house prices. Using data spanning three decades, we show that there was an exogenous mortgage credit expansion in the mid-1990s when automated underwriting software became widely available. Automated underwriting plausibly led to a mortgage credit expansion because it was more precise in identifying credit risks than human underwriters, allowing the GSEs to purchase a wider variety of loans without increasing their aggregate exposure to risk.

Many have argued that the 2000s housing boom resulted from an exogenous credit expansion to low-income borrowers. Yet after the 1990s, we find little change in the correlation between income and mortgage origination amounts, even though there was a large increase and subsequent decline in real house prices in the 2000s. As outlined in the new narrative on the 2000s boom, a more plausible explanation of the boom is a broad-based increase in the demand for housing, potentially caused by optimistic beliefs about future house price growth

and fueled, at least in part, by the increased use of private-label securitization and subprime loans. It is possible that financial innovation in the 1990s initiated this broad-based increase in demand. Whether it actually did so is a question we leave to future research.

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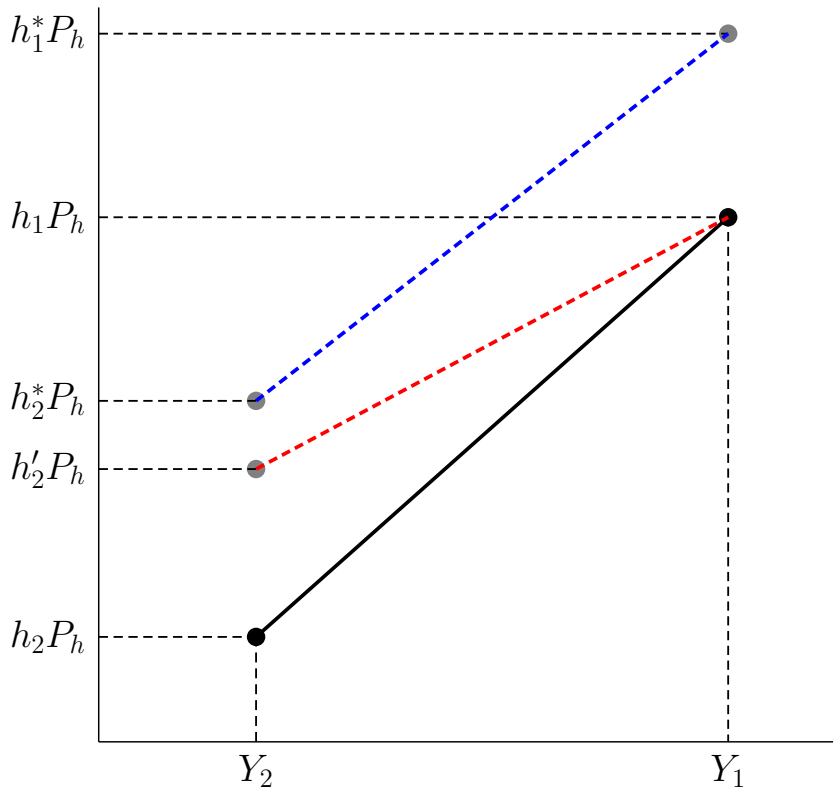

Figure 1. Theoretical Relationship Between Mortgage Debt and Income. *Note:* Graphical representation of theoretical model described in Section 2. The y-axis is spending on housing, which is the product of the price of a unit of housing (P_h) and the amount of housing (h). The x-axis is income. There are two types of households: low income (2) and high income (1). The black line represents the relationship between mortgage debt and income when low-income households are constrained and high-income households are not. If there is a relaxation in credit constraints, only low-income households will increase their consumption of housing as high-income households were already unconstrained. This is captured by the red dashed line. If there is a change in the user cost of housing (perhaps due to a change in interest rates or optimism about future house price growth) then both low- and high-income households increase their consumption of housing. This is captured by the blue dashed line.

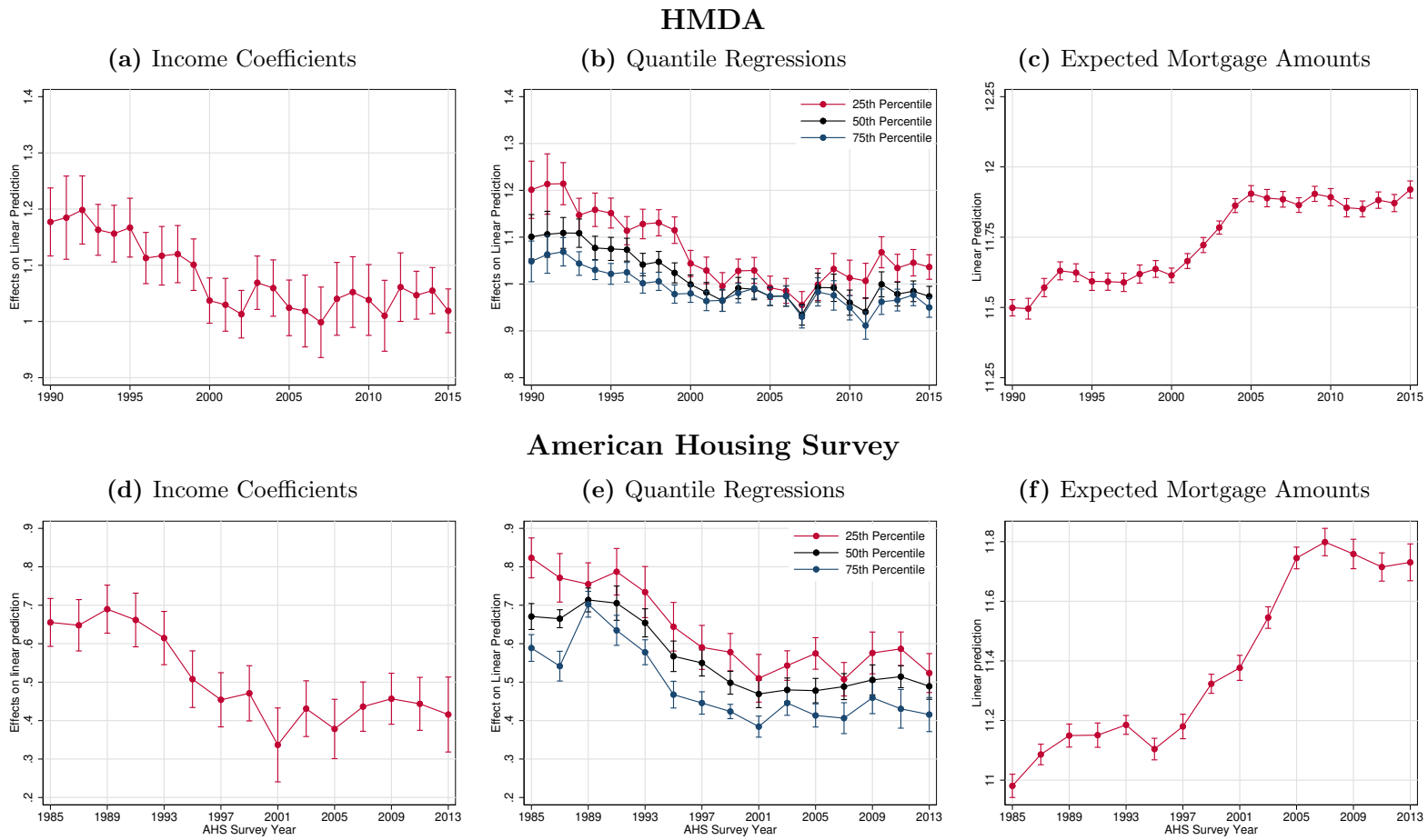

Figure 2. Regressions using Individual-Level Mortgage Balances and Income Levels.
Note: Figure 2a is a plot of the income coefficients from a regression of total individual purchase mortgage origination amounts on application income from HMDA, where income is instrumented with the most recently available Census tract income from the Census and ACS. The regressions include CBSA × year fixed effects and controls for the borrower's race and gender, all interacted with year. Figure 2b are parallel quantile regressions that they do not control for race and income. Figure 2c is a plot of expected mortgage amounts, which are predictions from an identical regression without CBSA fixed effects, holding income constant at its average value across all years. Figures 2d, 2e and 2f are graphs of income coefficients and expected mortgage amounts from comparable regressions using individual-level data from the AHS. The AHS regressions do not include CBSA × year fixed effects. All confidence intervals are 95 percent confidence intervals. HMDA regressions are clustered by CBSA. *Source:* Authors' calculations using data from the Home Mortgage Disclosure Act, US Decennial Census, American Housing Survey, and American Community Survey.

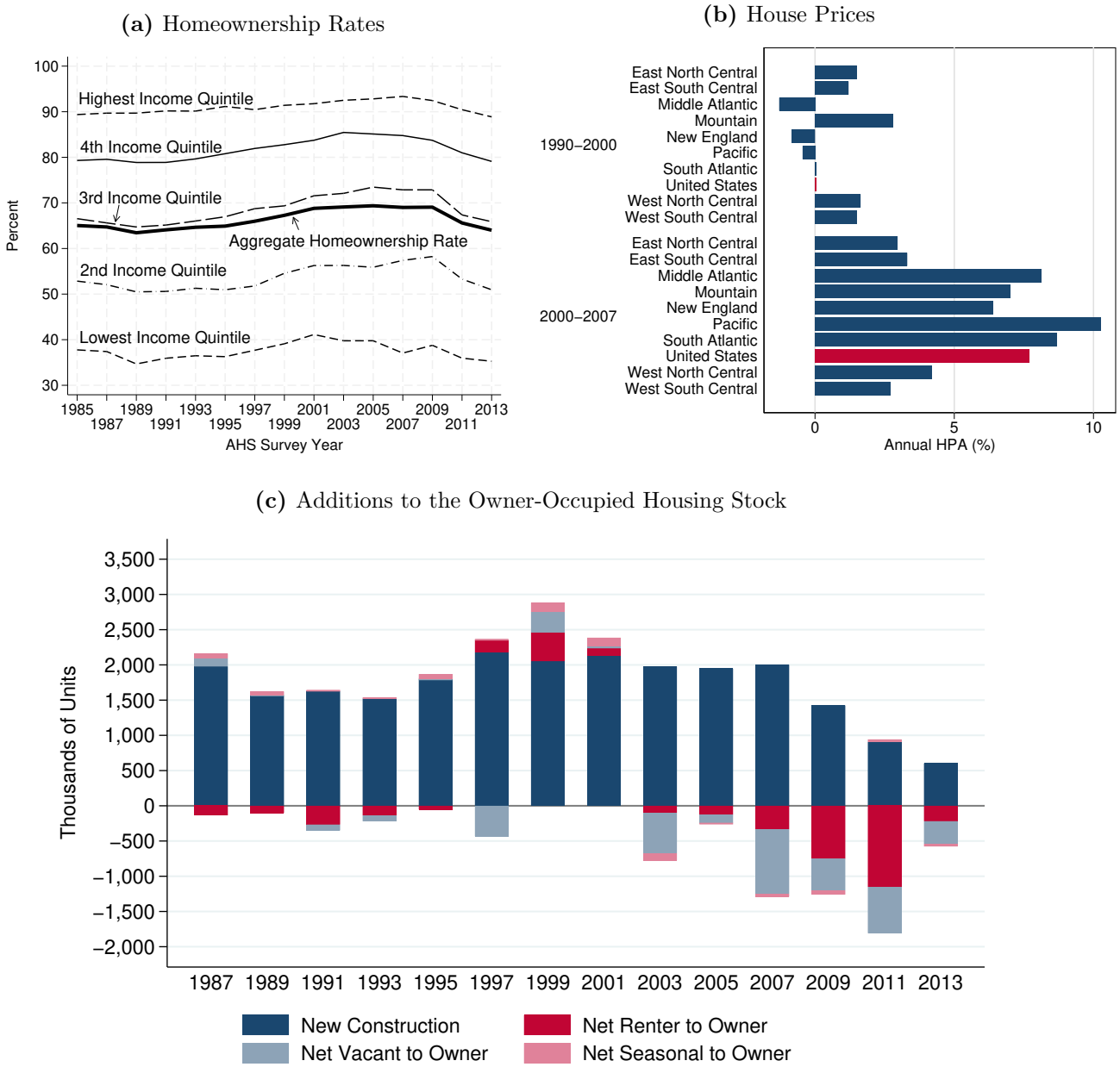

Figure 3. Implications of a Credit Expansion. *Note:* Homeownership rates in Figure 3a are constructed using microdata from the American Housing Survey (AHS) and correspond to households with heads aged 25–64 and positive incomes. Figure 3b depicts annualized house price growth across the specified years for US census divisions and the US as a whole using the CoreLogic house price index for single family homes, deflated with the headline CPI index excluding shelter. Figure 3c is constructed using microdata from the AHS.

Freddie Mac

GOLD MEASURE WORKSHEET—Version 2.0

Borrower/Co-borrower Name(s) _____
City, State _____
Lender Loan Number _____
Origination Date _____
Completion Date _____

Seller Name _____
Freddie Mac Seller Number _____
Freddie Mac Loan Number (if available) _____
Branch Office/Channel _____
Underwriting Center _____
TPO Name _____
Underwriter _____

Loan Decision Approved __ Denied __ Withdrawn __ File Closed __

Directions:
- Circle the appropriate "Risk Units" (RUs) for each category. Total the RUs in each section and enter on the Subtotal line. Then combine Subtotals for each section and enter the Grand Total on the Total RUs line. Note that negative numbers such as "–2" are risk offsets.
- It is important to read the accompanying *Gold Measure Worksheet and Instructions—Version 2.0* booklet and to refer to it for additional information on completing this worksheet.
- This worksheet is an aid, not a substitute for the underwriting decision.
- Complete **either** Credit File A **or** Credit File B, but not both. Use Credit File A if 3 credit scores are requested. Use Credit File B if fewer than 3 credit scores are requested. See the *Gold Measure Worksheet and Instructions—Version 2.0* booklet for easy instructions on how to order bureau and bankruptcy scores for use with Credit File A.

I. Credit File A

Directions: When using Credit File A, complete **either** the Bureau Score **or** the Bankruptcy Score, but not both.

Bureau Score Equifax Beacon Score, Trans Union Empirica Score and TRW-FICO Score (See instructions)	**RUs**	**Bankruptcy Score** Equifax DAS Score, Trans Union Delphi Score and TRW-MDS Score (See instructions)	**RUs**
Over 790	–16	150 or less	–12
771 – 790	–14	151 – 200	–10
761 – 770	–11	201 – 240	–4
731 – 760	–7	241 – 300	–3
721 – 730	–5	301 – 320	–1
701 – 720	0	321 – 360	0
681 – 700	6	361 – 420	4
661 – 680	8	421 – 480	8
641 – 660	12	481 – 540	11
621 – 640	17	541 – 620	15
601 – 620	20	621 – 700	18
581 – 600	23	701 – 740	21
541 – 580	25	741 – 840	23
540 or less	32	841 – 960	25
		Over 960	29
No reported Score available	20	No reported Score available	20

I. Credit File A.
Subtotal of circled **RUs**: _____

II. Income

	RUs
Self-employed and above area median income:	5
Majority of income from commissions:	5
Employed second earner on application:	–2
Borrower's time on job is 5 years or more:	–2
Co-borrower's time on job is 2 years or more:	–1

II. Income. **Subtotal** of circled **RUs**: _____

III. Loan, Collateral, Assets

LTV/TLTV (including secondary financing*) is:	**RUs**
60.5% or lower	–27
60.6 – 70.5%	–16
70.6 – 80.0%	–5
80.1 – 85.5%	–1
85.6 – 90.5%	0
90.6 – 93.5%	2
93.6 – 94.5%	5
94.6 – 95.5%	8
95.6 – 96.5%	10
96.6 – 98.5%	11
98.6 – 99.5%	13
99.6 – 99.9%	15

Property seller contributions exceed 3% of value: **RUs** 5

Reserves are:	**RUs**
Less than 1 month	8
At least 1, but less than 2 months	5
At least 2, but less than 4 months	0
At least 4, but less than 5 months	–3
5 or more months	–6

Less than 5% down from borrower funds with 95% LTV (e.g. Affordable Gold with 3/2 Option): **RUs** 8

III. Loan, Collateral, Assets.
Subtotal of circled **RUs**: _____

*When secondary financing is included, if the secondary financing provides for any amortization (payments) before maturity of the Freddie Mac loan, then add 1% to LTV for every rounded percentage point of secondary financing. Likewise, add 0.5% to LTV for every rounded percentage point of secondary financing, if there is no amortization (no payments due) before maturity of the Freddie Mac loan. Unsecured grants or gifts require no adjustments to LTV.

I. Credit File B

Directions: Use Credit File B if fewer than 3 credit scores are requested.

No delinquencies or other derogatory tradeline or derogatory public record information and number of tradelines (open or closed) is:	**RUs**
11 or more	–4
6 – 10	–3
1 – 5	0

One or more revolving tradelines and **total** revolving balances are under $500: **RUs** –4

Fewer than 3 tradelines (open or closed): **RUs** 2

Percent of all tradelines (open or closed) ever delinquent or worse (30-90 days or more, collection, charge-off, etc.):	**RUs**
0 – 10%	–3
11 – 15%	0
16 – 40%	4
41 – 60%	8
Over 60%	11

Worst ever derogatory credit file entry is either:	**RUs**
■ 30–180 days delinquent:	6
or	
■ Public record (bankruptcy, foreclosure, judgment, lien, garnishment, suit, certain collections) or tradeline reported as over 180 days delinquent, charge-off, repossession or collection:	10

Number of derogatory Public records:	**RUs**
0 – 1	0
2 – 3	4
Over 3	9

Number of inquiries in the past 3 months:	**RUs**
0	–2
1	0
2 – 3	5
4	8
5	11
More than 5	14

Age of **oldest tradeline** (in months):	**RUs**
0 (no tradelines) – 6	18
7 – 12	13
13 – 24	7
25 – 48	3
49 – 72	2
73 – 120	0
121 – 168	–1
169 or more	–2

If age of oldest tradeline is 13–48 months and **any one or more** of the following:
- More than 3 inquiries within the past 3 months
- More than 3 tradelines opened in the past year
- Total open balances exceed $10,000 **RUs** 6

I. Credit File B.
Subtotal of circled **RUs**: _____

IV. Debt-Payment Burden

Debt-to-income ratio is:	**RUs**
Less than 32.6%	0
32.6 – 38.5%	2
38.6 – 40.5%	4
40.6 – 42.5%	7
42.6 – 44.5%	10
44.6 – 46.5%	13
46.6 – 48.5%	15
48.6 – 50.5%	18
Over 50.5%	30

Spread between total debt and housing ratios (i.e. nonhousing debt ratio) is:	**RUs**
10 to 15%	2
More than 15%	5

Proposed housing expense is less than 120% of previous housing expense: **RUs** –1

IV. Debt-Payment Burden. **Subtotal** of circled **RUs**: _____

V. Loan/Property Type

Loan type is:		**RUs**
Fixed-Rate:	15-Year	–6
	20-Year	–4
	25-Year	–1
ARM:	Rate-Capped	6
	Payment-Capped	8

Property type is:	**RUs**
2 Unit	5
3–4 Unit	11
Condominium	5

V. Loan/Property Type.
Subtotal of circled **RUs**: _____

Total of sections I A or B, II, III, IV and V. **TOTAL RUs:**_____

Freddie Mac Risk Unit Guideline:	15 RUs
If pre-purchase counseling:	16 RUs
If post-purchase counseling:	17 RUs
If pre- and post-purchase counseling:	18 RUs

Refer to *Gold Measure Worksheet and Instructions—Version 2.0* booklet for more information. This worksheet is an aid, not a substitute for the underwriting decision. Call your Account Representative for additional information.

Figure 4. Freddie Mac's Gold Measure Worksheet
Source: Avery et al. (1996).

(a) Evaluating Alternative Loans Using the Gold Measure Worksheet

	LTV Ratio	FICO Score	DTI Ratio	Months of Reserves	Fixed or Adjustable Rate	Total Risk Units
Loan A	90%	Over 790	Over 50.5%	2–3	Fixed	
Risk Unit Increment	0	–16	+30	0	0	14
Loan B	90%	640	Below 32.6%	2–3	Fixed	
Risk Unit Increment	0	+17	0	0	0	17
Loan C	99.5%	Over 790	50.5%	5	Adj.	
Risk Unit Increment	+13	–16	+18	–6	+6	15
Maximum Number of Risk Units Suggested by Freddie Mac for Loan Acceptance						**15**

(b) Comparing Manual and Automated Underwriting on a Sample of Freddie Mac Affordable Gold Loans.

Method of Underwriting	**Acceptance Rate (%)**	**Relative Delinquency Rate (Accept)**	**Relative Delinquency Rate (Reject)**
Manual Underwriting	51.6	1.04	0.96
Gold Measure Worksheet	≈ 50	0.18	1.99
1995 Loan Prospector Model	44.8	0.21	1.63
2000 Loan Prospector Model	87.3	0.70	3.08

Table 1. Implications and Effectiveness of Automated Credit Evaluation. *Note:* The calculations in Table 1a correspond to single-family, 30-year mortgages for which no special cases apply (for example, the borrower is not self-employed). The DTI ratio is the back-end ratio, which encompasses other monthly payments in addition to mortgage payments. The adjustable-rate mortgage in Loan C is a rate-capped ARM (not a payment-capped ARM). In Table 1b, the default rates are 90-day default rates within 3 years of origination expressed relative to the average default rate of the overall sample. *Source:* Table 1a is constructed by the authors using the Gold Measure Worksheet in Figure 4. For Table 1b, the numbers for the manual underwriters and the 1995 and 2000 Loan Prospector Models are from Gates et al. (2002). Numbers for the Gold Measure Worksheet are backed out using data in Table 2 of Straka (2000).

A Data Appendix

A.1 HMDA

HMDA includes information collected from mortgage originators on nearly all mortgage originations in the US from 1990 until the present. Information collected includes the dollar amount of mortgage; the purpose of the mortgage (refinance, purchase, or home improvement); the date of the mortgage origination; the census tract of the underlying property; and information on the borrower including the borrower's application income, race, gender, and whether the borrower is an owner-occupant. Unless otherwise noted, we only consider owner-occupied properties with non-missing race and gender information.[15]

During the 2000s housing boom, first-lien mortgages were often supplemented by second liens at purchase ("piggyback loans"), so it is important to account for second liens consistently when making comparisons that span several years. While information is collected on first and second liens, the lien type is only reported starting in 2004.[16] We therefore created an algorithm to identify second liens made before 2004, and then validate this rule using the reported liens available starting in 2004.[17] Even when second liens are identified in HMDA, they are not matched to their corresponding first liens. Our algorithm both identifies second liens and matches them to their first liens. We then use the sum total of the first and second liens for each borrower in our loan-level regressions.

Figure A.1 depicts some summary statistics from data on purchase-mortgage originations in HMDA. The top panel shows the total number of purchase loans originated, along with the fraction of those loans that have associated second liens. The number of purchase loans originated increases steadily from 1990 until 2005, while the use of piggyback loans grows dramatically in the mid-2000s, near the end of the housing boom. After 2007, both series drop sharply. The bottom panel displays the median total loan amount for owner-occupied purchase mortgages, with this total encompassing both first liens and any associated piggybacks. The panel also shows the share of purchase mortgages made for owner-occupied properties, which declines by almost 10 percentage points during the boom. Note also that there is a hump shape in the median loan amount at the peak of the boom, when the piggyback share and the investor share were highest.

[15] Avery et al. (2007) suggests that missing race and gender indicates a business loan as opposed to a persona mortgage.

[16] A summary of all changes made to HMDA in 2004 can be found in "HMDA Changes Are On the Way; New Rules Take Effect in 2004," which appears in *Community Dividend*, an online publication of the Federal Reserve Bank of Minneapolis, available at https://www.minneapolisfed.org/publications/community-dividend/hmda-changes-are-on-the-way-new-rules-take-effect-in-2004.

[17] Details of the second-liens algorithm appear in Appendix A.1.1.

A.1.1 Identifying Piggyback Loans in HMDA

The procedure for identifying piggyback loans in HMDA involves multiple stages. Each stage involves the identification of some piggyback loans, and after each stage, those piggyback loans and their associated first liens are removed from consideration.

The first stage uses loan-level characteristics available in HMDA to identify duplicate observations. Of the two observations that comprise a duplicate observation, we assume that the larger loan value is the first lien and that the smaller is the piggyback loan. The loan-level characteristics that we used to identify these duplicates are as follows:

1. The banking institution originating the loan.
2. The week that the loan was originated.
3. The month that the loan was originated.
4. The Census tract in which the property is located.
5. The loan type. Loans can be either conventional or guaranteed by the Federal Housing Administration (FHA), the Veterans Administration (VA), the Farm Service Agency (FSA) or the Rural Housing Service (RHS).
6. Whether or not the borrower will be occupying the property.
7. The income, race, and sex of the borrower.

The second identification stage is similar to the first, but in place of some of the loan-level characteristics, this stage assumes specific ratios of the origination amounts of first and second liens. First, we assume that the first lien is four times the value of the second lien (an 80–20 ratio), which is the most common ratio of first liens to their associated piggybacks. Then we assume that the ratio is 80–10, 80–5, and finally 80–15, which are also common ratios observed in the data. The loan-level characteristics used to match loans in this stage are the month of origination, the property's Census tract, and the borrower's income.

The third stage is more ad hoc and identifies piggyback loans using duplicate observations in a variety of loan-level variables:

1. The origination date; the application date; the property's Census tract; the borrower's income; and whether the borrower intends to occupy the property.
2. The origination date; the application date; the property's Census tract; the income of the borrower rounded to the nearest $10,000; whether the borrower plans to occupy the property; the borrower's sex and race; the co-applicant's sex and race; and whether the loans are conventional or guaranteed by one of the federal authorities listed above.

3. The origination date; the property's Census tract; whether the borrower is an owner-occupant; the banking institution that made the loans; and the income, race, and sex of the borrower.

4. The origination date; the banking institution that made the loans; the property's Census tract; whether the borrower intends to occupy the property; whether the loans are conventional or guaranteed by one of the federal authorities listed above; the race, sex, and income of the borrower; and the race and sex of the co-applicant.

5. The origination date; the banking institution that made the loans; the week the loans were applied for; the property's Census tract; whether the borrower intends to occupy the property; whether the loans are conventional or guaranteed by one of the federal authorities listed above; the income of the borrower rounded to the nearest $5,000; and the race and sex of the borrower and co-borrower.

6. The origination date; the week of application; the property's Census tract; whether the borrower intends to occupy the property; whether the loans are conventional or guaranteed by one of the federal authorities listed above; the income, race, and sex of the borrower; and the race of the co-borrower.

As a last step, we use county-level median house prices from Zillow, and code any loan that is less than 0.01 percent of the value of the median house price in the county as a piggyback loan. We are not able to identify the first liens associated with these piggyback loans, so they are removed from the analysis.

Starting in 2004, HMDA identifies the lien type of all reported loans, but does not link associated first and second liens. To assess the quality of our identification of piggyback loans, we compare our results to the HMDA-identified second liens in 2004. Of the loans that HMDA denotes as first liens, we identify more than 99 percent of them correctly as first liens. Of the loans that are coded in HMDA as second liens, we correctly identify 84.8 percent as second liens. Therefore, we falsely identify 15.2 percent of HMDA-coded second liens as first liens, and 0.9 percent of HMDA-coded first liens as second liens. Because there are many more first liens than piggyback loans, the number of falsely identified first liens is a little under half the number of falsely identified second liens. Unfortunately, these errors do not cancel each other out, but overall we correctly identify 97 percent of the loans. This rate holds for all years after 2004 as well. We also checked to see if our procedure was more accurate in higher or lower income areas. Using 2000 Census tract income, in 2004 we correctly identify 97.8 percent of the loans in the lowest quartile of Census tracts by median income, and 96.9 percent of the loans in the highest quartile of Census tracts.

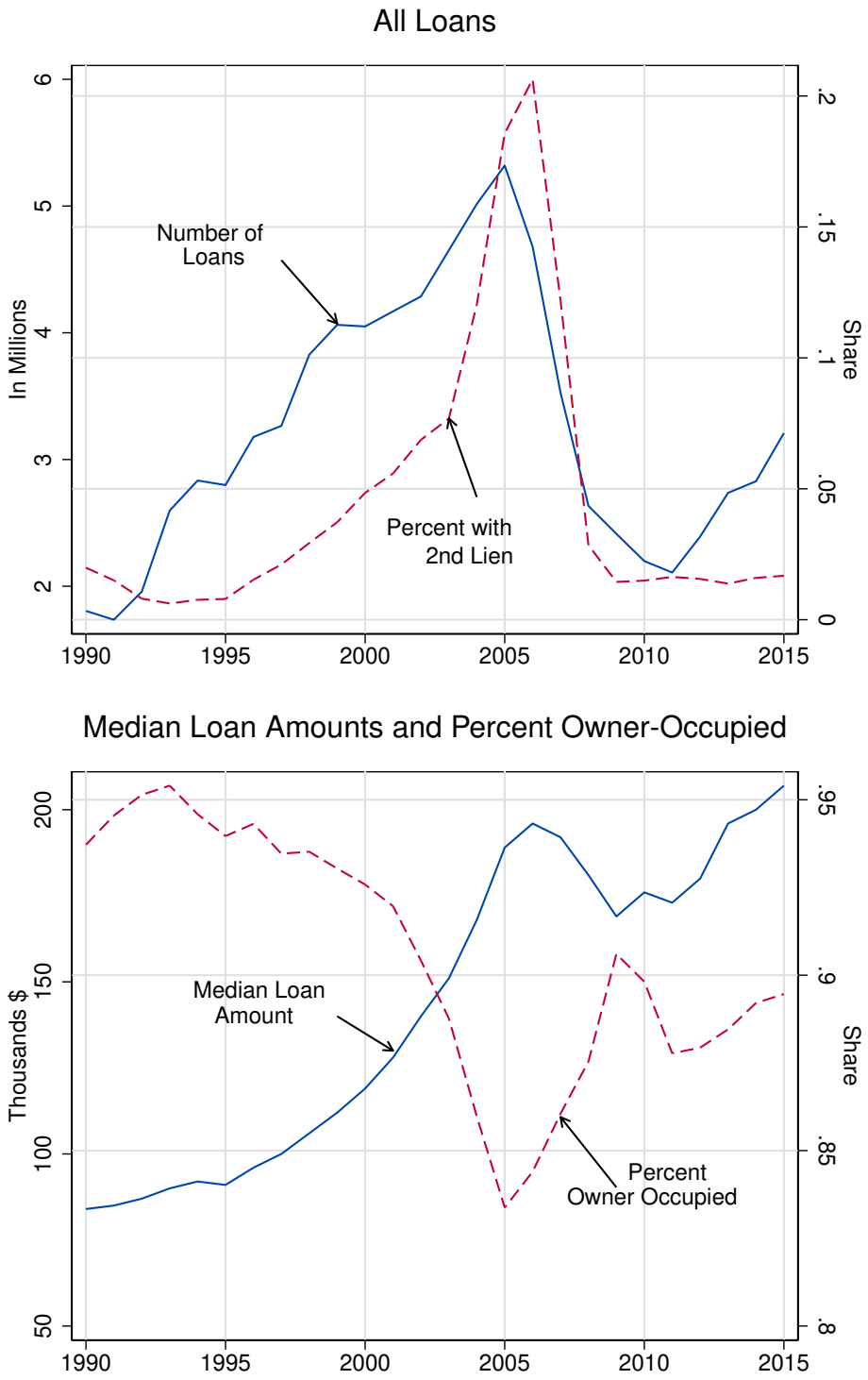

Figure A.1. Summary Statistics from HMDA. *Note:* The total median loan amount refers to the median value of the sum of first liens and associated piggyback loans on owner-occupied properties. The percent with second lien is the share of all loans that have a piggyback loan. *Source:* Authors' calculations using microdata from the Home Mortgage Disclosure Act (HMDA).

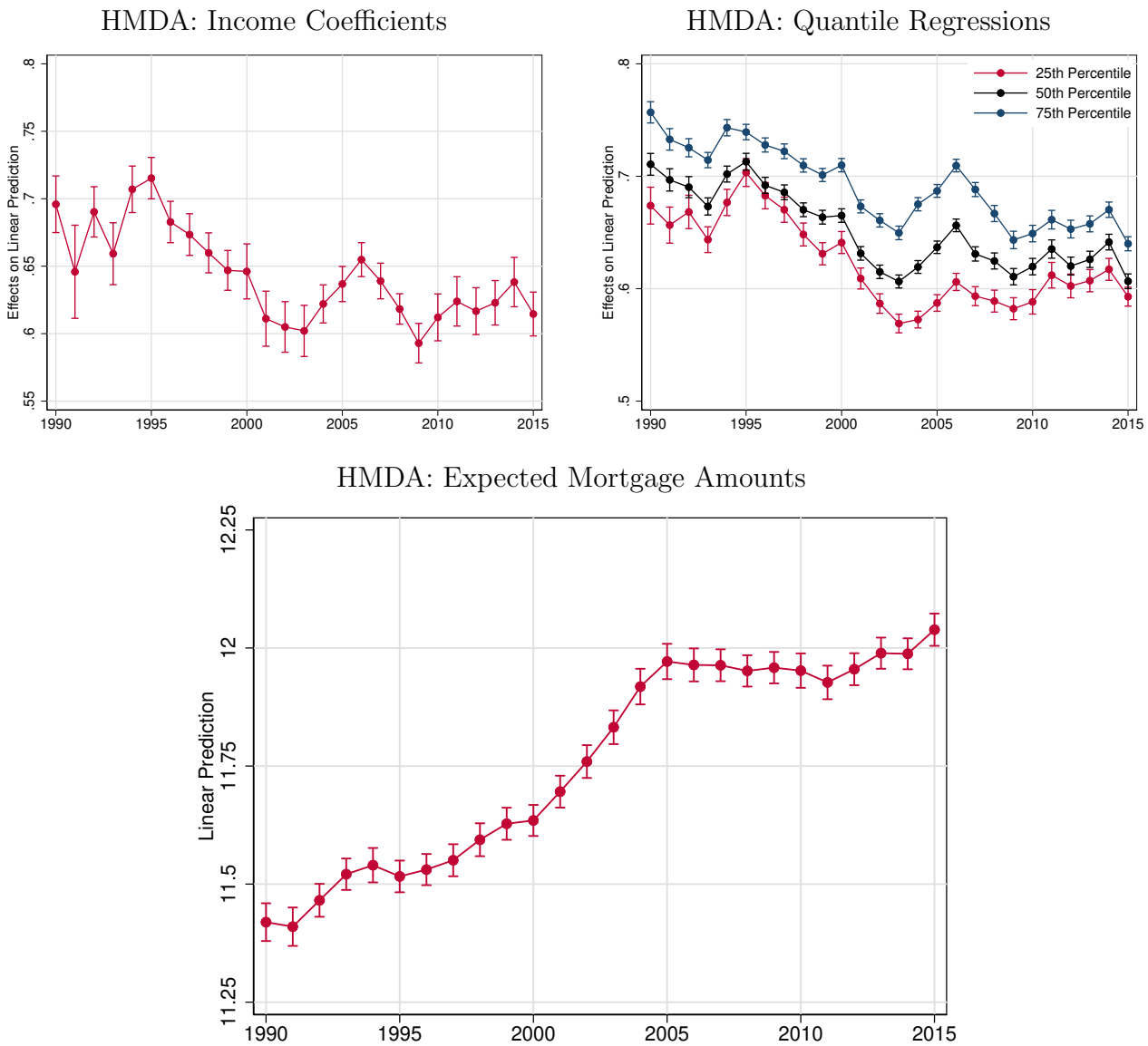

Figure A.2. Non-IV Regressions Using HMDA. *Note:* These are parallel results to Figures 2a, 2b and 2c, but without instrumenting for income. *Source:* Authors' calculations using HMDA.

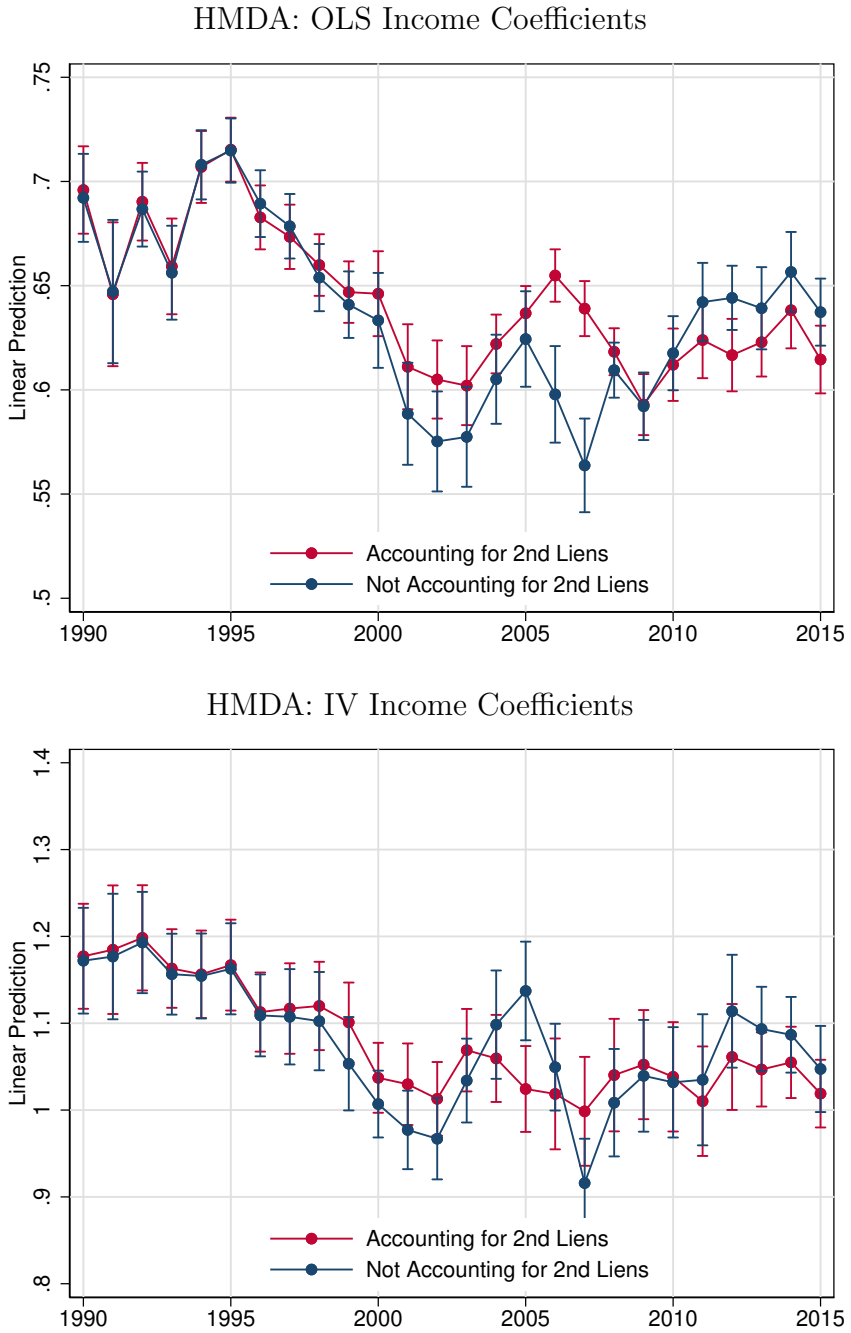

Figure A.3. Regression Results With and Without Accounting for Second Liens in HMDA. *Note:* These figures illustrate the effects of accounting for second liens in HMDA on the coefficient on income in Equation 9. Not accounting for second liens includes piggyback purchase loans as separate observations. *Source:* Authors' calculations using HMDA.

A.2 American Housing Survey

Our second dataset is the American Housing Survey (AHS), a joint project of the US Department of Housing and Urban Development (HUD) and the Census Bureau. The AHS began in 1973 and has been conducted in every odd-numbered year since 1981. It is intended to measure the size, quality, and composition of the US housing stock, and also includes information on the housing costs, demographic characteristics, and incomes of sampled households.[18] The survey is longitudinal, with interviewers returning to the same housing units in each survey. Units are added to the AHS sample though new construction and conversions from non-residential use, and units are removed when they are no longer inhabitable. A new sample of housing units was drawn for the 1985 survey and retired after the 2013 survey, so our dataset includes every odd-numbered year from 1985 through 2013.

A.2.1 Mortgage Debt and Income for Recent Home Purchasers

The AHS can be used to analyze the flow of new purchase-mortgage debt because it includes information about the size of any existing mortgages at the time those mortgages were originated. Our regression sample consists of homeowners who (i) moved into their residences since the last AHS survey, and (ii) are currently carrying a mortgage that was originated in the same year that their unit was purchased. Table A.1 illustrates the construction of our sample. The first column displays the total number of observations in each AHS survey, while the second column includes the number of interviews that correspond to occupied units.[19] The third column shows the number of occupied interviews corresponding to homeowners, and the fourth shows the number of homeowners that moved into their units since the last AHS survey. The last column shows the number of recently moving homeowners with debt. This column comprises our regression sample, which totals about 39,000 households across all survey years.

The income measure used in the AHS regressions is family income, an expansive measure that includes regular income sources such as wage and business income, but excludes capital gains. Because capital gains are not expected to continue each year, they are appropriately excluded from income that would help a family qualify for a mortgage. Data on income components and mortgage debt are topcoded in early years of our sample. We account for topcoding of the most important income component (wage income) and mortgage debt following the common practice of using a model to estimate a mean value above the topcode,

[18] For a general description of the AHS, see US Department of Housing and Urban Development (2017). For discussions of the AHS as a data source for housing-finance issues, including new mortgage debt, see Lam and Kaul (2003), Eggers (2007), and Eggers (2008).

[19] In addition to occupied units, the AHS also collects information on vacant units and on households that are currently occupying a sampled unit but usually live somewhere else (for example, a household occupying a vacation property).

and then assigning all topcoded values that mean value.[20] In later years of the sample, the Census Bureau essentially performs this correction for us, by replacing values above a topcode with the mean of actual data above the topcode.[21]

Table A.2 depicts summary statistics for the AHS regression sample. Both weighted and unweighted statistics are reported; our regressions use sample weights. In unreported work, we found broad agreement between the average age of new home buyers in the AHS data and separate age-at-purchase data compiled by the National Association of Realtors.

For mortgage debt, we include additional mortgages beyond the first mortgage for households that report them. AHS debt is therefore comparable to debt in HMDA, which includes second liens. The top panel of Figure A.4 compares median levels of new mortgage debt from HMDA and the AHS. The debt estimates are similar in the two datasets, especially after subsetting on metropolitan areas, where HMDA reporting is concentrated. Less agreement is seen in the lower panel of the figure, which compares AHS and HMDA incomes. In particular, HMDA income rises above AHS income during the height of the 2000s housing boom. This pattern confirms a lesson from Avery et al. (2012), who note that income reported to HMDA potentially overstates a borrower's true income in areas or periods where house prices are rising rapidly. The data are also consistent with Blackburn and Vermilyea (2012), who use the AHS to measure HMDA income overstatement from 1995 to 2007. The authors compare the incomes of individual home purchasers in the AHS to local averages of reported HMDA incomes for similarly sized mortgages that have matching borrower and loan characteristics. They find income overstatement of around 15–20 percent in 2005 and 2006.

A.2.2 Flows of Properties Across Various Forms of Tenure

The longitudinal nature of the AHS allows us to estimate the number of units that transition from one type of tenure to another between survey years. Similar estimates are reported by HUD using AHS data in its Components of Inventory Change (CINCH) reports.[22] These reports are intended to describe how the US housing stock evolves. For example, they include estimates of new construction, demolitions, and conversions from non-residential use. They also include estimates of the number of units that transition from one tenure type to another. Unfortunately, the CINCH transition estimates are not precise enough for our purposes, because they correspond to transitions from one tenure type to *any other* tenure type. For example, CINCH data indicate the number of rental properties that transition to either owner-occupancy or vacancy, but not the number that transition to owner-occupancy

[20] We use a Pareto distribution to estimate the above-topcode mean.

[21] The Census Bureau begins this method of reporting for wage income in the 1999 AHS and for debt in the 2005 AHS.

[22] For information on CINCH reports, see https://www.huduser.gov/portal/datasets/cinch.html.

specifically. Also, CINCH estimates of the number of new owner-occupied units added through construction at the end of the 1990s appear to be inaccurate, as there is a puzzling dip in this number in the 1997–1999 CINCH report.[23]

Because of this likely error and our need to calculate flows across specific forms of tenure, we created our own CINCH-like dataset. The longitudinal weights in the dataset allow us to estimate the total number of specific tenure changes and as well as additions to the stocks from new construction. The weights are constructed by following descriptions of CINCH weighting procedures in Eggers and Moumen (2005). As a check on the algorithm, Figure A.5 compares the total number of newly constructed units implied by our AHS microdata to estimates of total housing completions published by the Census Bureau. There is substantial agreement between the two series. The dip in total new construction found in the CINCH data at the end of the 1990s is not present in our series. The dip is also absent in the dark blue bars of Figure 3c of the main text, which show new construction for owner-occupied homes specifically.

A.2.3 Comparison of AHS Homeownership Rates with Official Rates

In Figure 3a we use AHS microdata to calculate homeownership rates for households with heads aged 25–64 and positive incomes. Figure A.6 compares homeownership rates calculated with AHS data for all households to official US homeownership rates, which are generated from the Current Population Survey/Housing Vacancy Survey (CPS/HVS) and published by the Census. The figure shows that the AHS and CPS/HVS homeownership rates are extremely close for all years except 2009.

[23]The 1997–1999 CINCH report states that 715 thousand new owner-occupied units were created via new construction between these two AHS surveys. This is less than half the corresponding number in either the 1995–1997 CINCH report (2.7 million units) or the 1999–2001 report (2.5 million units). The 1997–1999 dip appears inconsistent with the relatively stable number of total housing completions in aggregate Census data.

Survey Year	AHS Observations	Occupied Interviews	Occupied Interviews: Homeowners	Recently Moving Owners	Recent Movers w/Debt
1985	48,830	36,952	24,082	3,943	2,872
1987	49,641	43,104	28,693	3,981	2,735
1989	51,823	38,957	25,337	2,957	2,088
1991	51,027	44,193	29,322	3,559	2,437
1993	55,981	40,317	26,098	3,064	2,084
1995	51,734	44,753	28,841	4,232	3,420
1997	46,010	39,021	25,833	3,529	2,623
1999	52,912	45,459	30,247	4,167	3,167
2001	48,347	41,277	28,099	3,892	2,790
2003	55,593	46,970	31,996	4,430	3,115
2005	50,611	42,576	29,272	4,394	3,306
2007	46,555	38,434	26,411	3,322	2,559
2009	53,448	44,133	29,879	2,831	2,115
2011	61,003	50,693	30,241	2,469	1,842
2013	70,044	58,274	35,229	2,855	2,012
Totals	793,559	655,113	429,580	53,625	39,165

Table A.1. Sample Selection from the American Housing Survey. *Note:* The first column lists the total number of observations in the AHS by survey year, and the second lists the number of AHS interviews that could be collected from occupied residences. The second column excludes observations with zero AHS sample weights or non-positive incomes. The remaining columns refer to occupied interviews of homeowners. The third column lists the total number of homeowner interviews, and the fourth refers to owners who moved into their homes during the previous two years. The last column lists the number of recently moving owners with mortgage debt. This column comprises our regression sample. *Source:* Authors' calculations using AHS microdata.

Panel A: Unweighted								
Survey Year	No. of Obs.	Median Income	Median Debt	Mean Income	Mean Debt	Mean Age	Pct. White	Pct. Female
1985	2,872	34,575	50,000	40,560	56,153	36.0	88.3	16.1
1987	2,735	38,000	59,000	45,669	66,195	36.9	87.6	16.9
1989	2,088	41,300	64,750	50,876	80,815	37.3	85.7	17.6
1991	2,437	45,000	69,000	55,545	86,286	37.7	86.3	21.7
1993	2,084	45,504	74,000	56,631	87,568	37.9	84.6	25.1
1995	3,420	48,660	79,000	60,542	93,958	38.5	80.4	28.5
1997	2,623	50,070	83,000	64,363	97,244	38.8	81.3	34.6
1999	3,167	57,000	100,000	72,792	115,946	39.6	78.3	38.5
2001	2,790	60,000	108,000	84,027	125,907	39.2	76.9	38.7
2003	3,115	60,000	125,000	79,931	150,013	39.4	75.5	40.3
2005	3,306	64,782	147,000	79,691	187,489	39.6	75.0	41.3
2007	2,559	68,000	160,000	88,188	221,028	40.2	74.6	41.3
2009	2,115	70,100	159,000	88,352	202,189	40.0	76.0	43.7
2011	1,842	67,964	152,000	90,422	191,821	40.0	73.3	43.9
2013	2,012	74,987	160,000	100,424	200,487	40.9	74.6	40.3
Total	39,165							

Panel B: Weighted								
Survey Year	No. of Obs.	Median Income	Median Debt	Mean Income	Mean Debt	Mean Age	Pct. White	Pct. Female
1985	2,872	35,000	50,000	40,888	56,801	36.1	87.9	16.0
1987	2,735	38,300	60,000	46,008	66,929	37.0	86.5	17.2
1989	2,088	42,000	66,000	52,080	82,815	37.4	85.6	18.1
1991	2,437	44,884	69,000	55,470	86,534	37.6	85.4	22.6
1993	2,084	45,500	73,500	56,350	87,165	38.0	84.6	25.5
1995	3,420	47,450	76,000	59,424	90,406	38.4	81.5	28.3
1997	2,623	50,000	81,000	63,696	95,600	38.7	80.6	35.1
1999	3,167	55,000	95,500	70,669	110,753	39.3	78.7	38.9
2001	2,790	59,000	103,600	82,700	122,680	39.0	77.3	39.6
2003	3,115	60,000	120,000	77,691	145,088	39.1	76.7	41.1
2005	3,306	64,000	145,000	78,722	186,253	39.1	75.2	41.8
2007	2,559	67,000	160,000	86,937	220,844	39.6	74.9	42.0
2009	2,115	67,000	150,000	84,662	196,459	39.0	75.9	44.6
2011	1,842	64,782	148,000	87,161	184,290	39.2	75.4	44.9
2013	2,012	72,700	150,000	95,247	194,029	39.9	74.6	40.3
Total	39,165							

Table A.2. Summary Statistics from the American Housing Survey. *Note:* Income and debt are topcode-adjusted. All observation counts are unweighted. Demographic means in the last three columns refer to the reference person of the household. *Source:* Authors' calculations using AHS microdata.

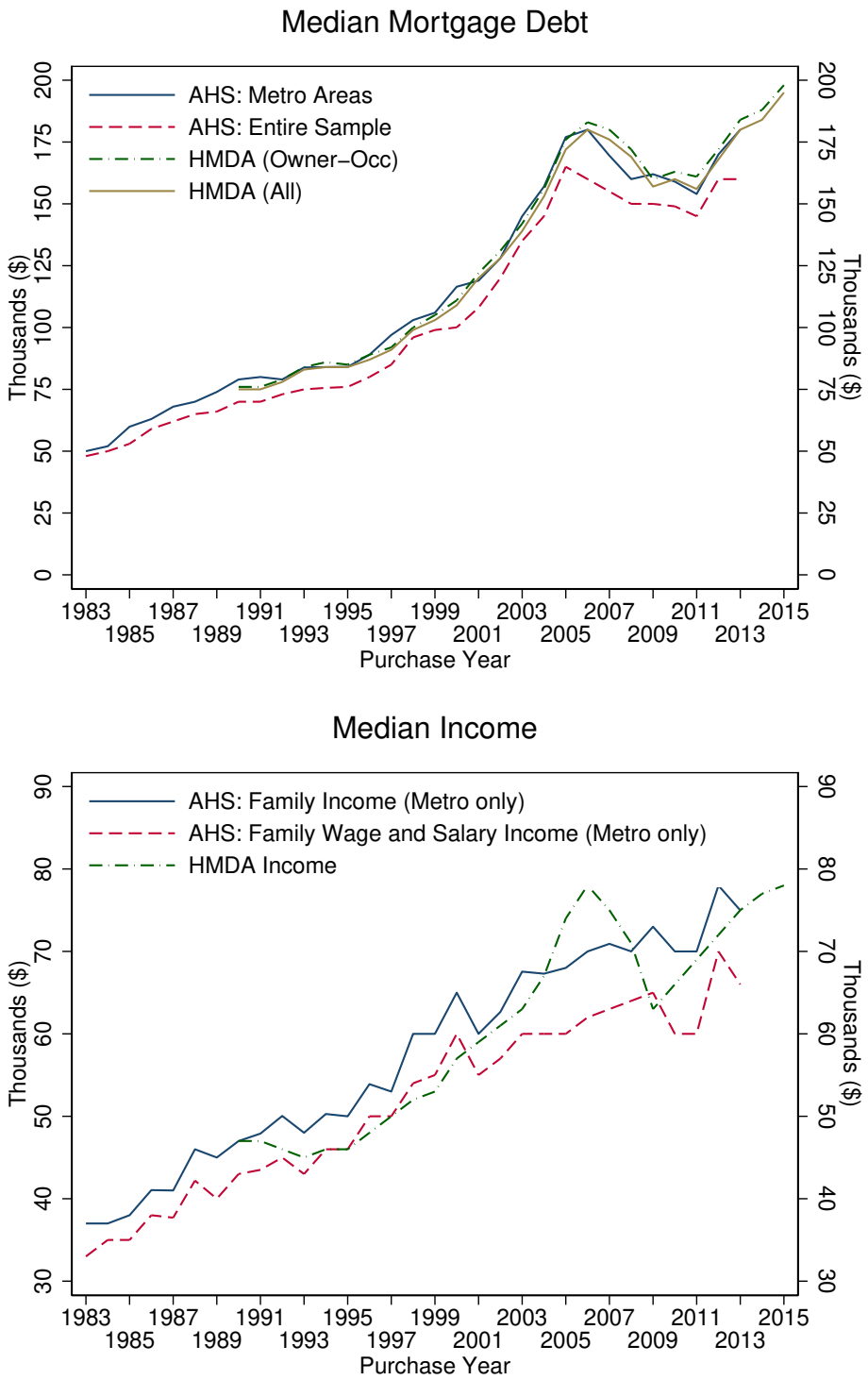

Figure A.4. Comparison of Median New Mortgage Debt and Median Income from the Home Mortgage Disclosure Act and the American Housing Survey. *Source:* Microdata from the Home Mortgage Disclosure Act and the American Housing Survey.

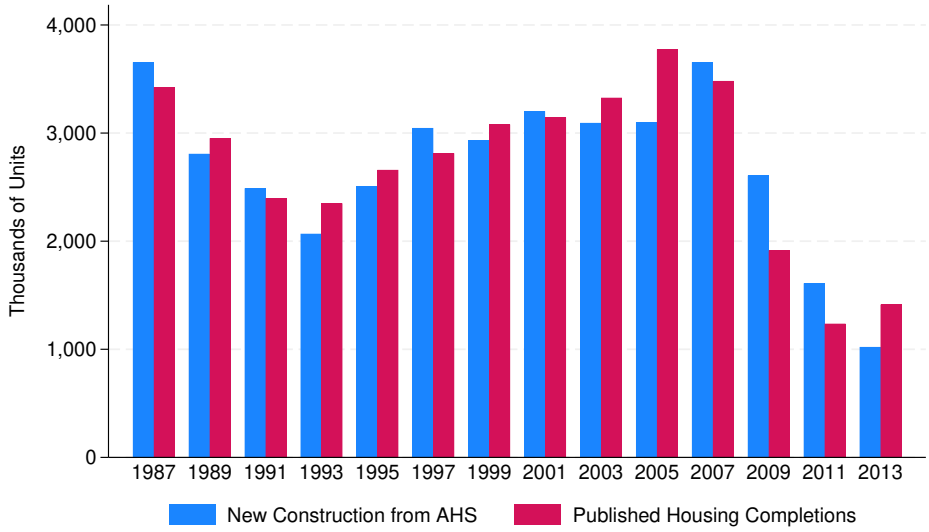

Figure A.5. Implied New Construction in AHS Microdata and Published New Completions: 2-Year Basis. *Note:* The blue bars depict total new housing-unit completions implied by AHS microdata in the given year and the immediately preceding year. The red bars denote published new completions from the Census Bureau over the same two years. *Source:* Census Bureau and authors' calculations using AHS microdata.

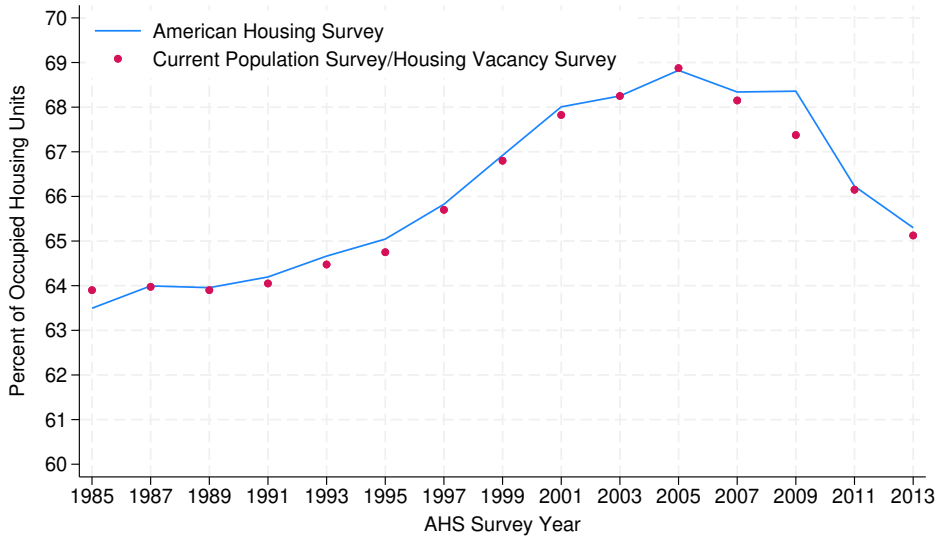

Figure A.6. Comparison of AHS Homeownership Rates with Official Rates from the Current Population Survey/Housing Vacancy Survey. *Note:* The red dots correspond to the official US homeownership rate, which is generated by the Current Population Survey/Housing Vacancy Survey and published by the Census Bureau. The blue line corresponds to the aggregate homeownership rate among AHS respondents in occupied housing units. The AHS data used for Figure 3a in the main text subsets on households with positive incomes and household heads aged 25–64. *Source:* Census Bureau and authors' calculations using AHS microdata.

B Comparison with Mian and Sufi (2009)

In this section, we briefly expand on the discussion of Mian and Sufi (2009) in Section 3.3. Mian and Sufi (2009) report that "2002-–2005 is the only period in the past eighteen years in which mortgage growth and income growth are negatively correlated. In all other time periods, income growth and mortgage growth are positively correlated, as one would expect under *standard models* of mortgage lending" (p. 1453, emphasis added). By "standard models," they likely mean the basic expectation that mortgage debt should be positively related to borrower income, because income determines ability to repay. We find little change in that relationship during the early 2000s and the relationship is strongly positive in all the years we study; only during the 1990s does the relationship change. What accounts for the difference between our findings and their finding of a negative correlation during the 2000s boom?

The first issue concerns what Mian and Sufi (2009) measure. Standard models of mortgage lending describe the relationship between individual loan amounts and individual borrower incomes. However, rather than examining individual mortgage amounts, they analyze the growth in the total value of all new purchase mortgages in a zip code. As we discuss in Section 3.3, this aggregation means their results are not really comparable with ours.

The second issue concerns the specification of their regression. Our main estimation equation (9) relates a household's *level* of new mortgage debt to its *level* of income. Mian and Sufi (2009) analyze growth rates—log first differences—instead:

$$\Delta D_{it} = \gamma_t \cdot \Delta I_{it} + u_{it}, \tag{10}$$

where the i subscript now represents zip codes (not households) and where we omit controls and fixed effects for simplicity. This equation appears to be a straightforward first-differenced version of our main estimating equation (9), but it contains a critical flaw. Omitting fixed effects and controls, our levels equation is $D_{it} = \beta_t I_{it} + \epsilon_{it}$. When β_t is time varying, the correct first-differenced specification requires an additional term:

$$\Delta D_{it} = (\beta_t - \beta_{t-1}) \cdot I_{it} + \beta_t \cdot \Delta I_{it} + \Delta\epsilon_t. \tag{11}$$

By omitting the levels term $(\beta_t - \beta_{t-1}) \cdot I_{it}$, the Mian and Sufi (2009) specification suffers from omitted variable bias, so their estimate of γ_t is unlikely to replicate the estimate of β_t. The differenced specification is valid only when β_t remains constant—that is, when credit conditions do not change over time ($\beta_t = \beta_{t-1}$). But that assumption contradicts the very phenomenon they seek to document.

An additional problem with the first-difference specification is that it throws away in-

formation, resulting in unstable estimates that are attenuated towards zero and may be negative in some cases. This point is illustrated by Figure B.7, which puts Mian and Sufi's negative correlation finding into context. Each panel in this figure is a binned scatter plot of zip-code level data corresponding to a specific time period. In the left panels, the vertical axes plot differences in average mortgage amounts across the beginning and end years of the period; differences in average incomes are plotted along the horizontal axes.[24] The right panels plot average levels of mortgage amounts and incomes instead. All mortgage and income data are normalized by dividing though by the period-specific standard deviation of that variable's differences (left panels) or levels (right panels). In each panel, the vertical axes span 4 standard deviations, while the horizontal axes span 8 standard deviations. It is therefore easy to compare the effect of a one-standard-deviation increase of income on debt in differences to the effect of a comparable increase in levels.

The clear lesson from the figure is that in levels, the relationship between income and debt is always positive and quite stable during the early 2000s, as we demonstrated in our empirical work. Yet the first-difference approach consistently yields very low increases in debt (growth) given an increase in income (growth). If taken literally, the differenced results would suggest that the standard models have *never* been a meaningful description of mortgage lending—a conclusion sharply at odds with the robust positive cross-sectional correlations in levels that we observe.

The discrepancy between first-difference and levels analyses occurs because standard underwriting models describe an inherently cross-sectional relationship between incomes and loan amounts at the borrower level. Mian and Sufi attempt to uncover this relationship using zip-code level changes in incomes and debt over time. But transitory income fluctuations and measurement error in both average loan amounts and in average household incomes measured at the zip-code level attenuate the first-difference correlation. As a result, the differenced specifications miss the true underlying relationship between income and debt relevant for lending and borrowing decisions.

[24]To be comparable with Mian and Sufi (2009) the income data are zip-code level average incomes from the Statistics of Income published by the Internal Revenue Service (IRS).

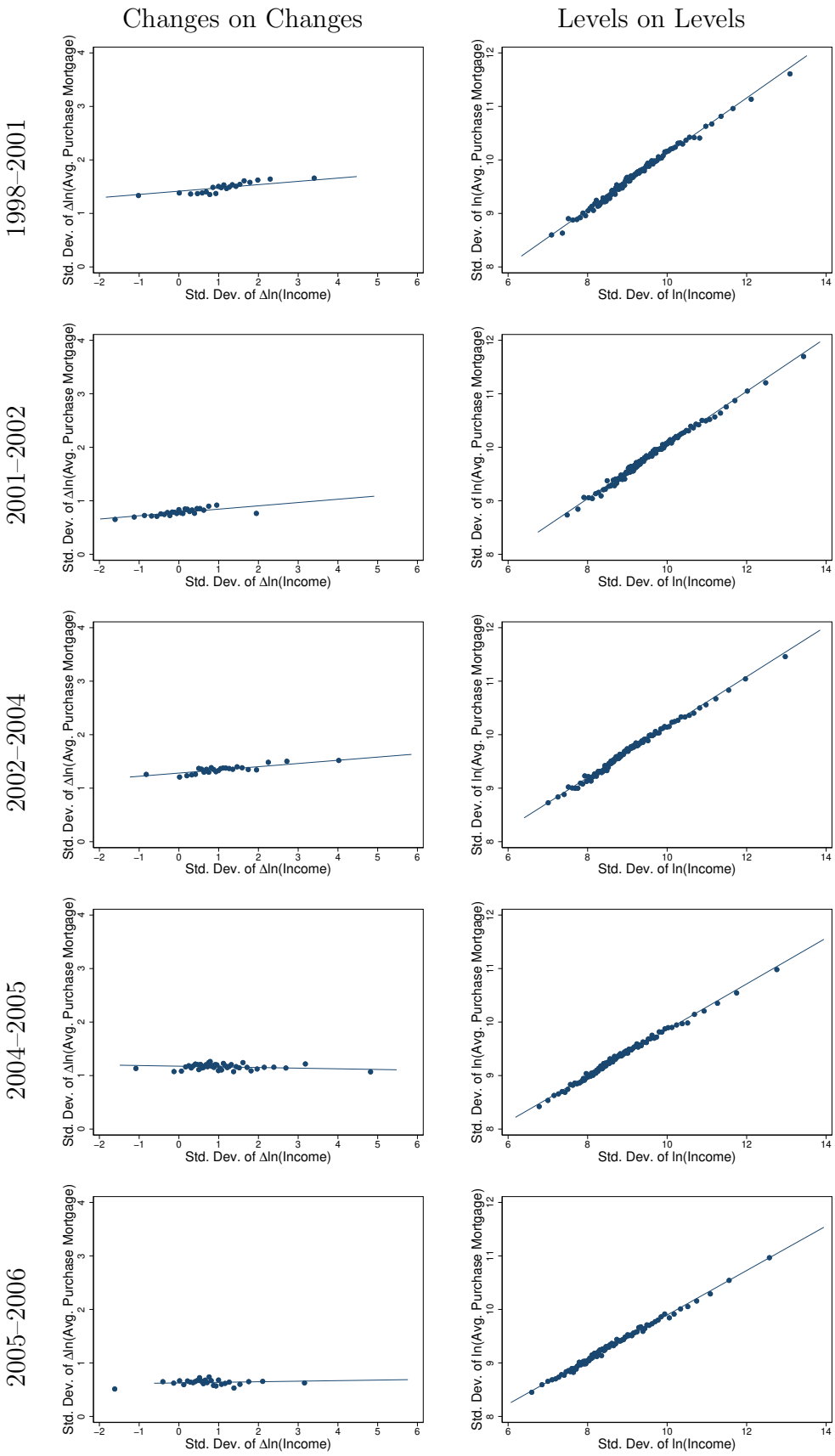

Figure B.7. The "Negative Correlation" between Mortgage Debt Growth and Income Growth in Differenced Data. *Note:* All figures are binned scatter plots of data after demeaning from county means and normalizing by period-specific standard deviations. All zip codes include at least 4,000 tax returns to approximate the sample sizes used in Table 2 of Adelino et al. (2016) and Table 8 of Mian and Sufi (2017). *Source:* Authors' calculations using IRS Statistics of Income and HMDA.